DATA SCIENCE LABORATORY MANUAL

Module III ? Data Visualization

Topic: Charts and Graphs

OVERVIEW & LEARNING OBJECTIVES

Selecting the right visualization for categorical, numerical, and time-series variables.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Charts and Graphs in modern Data Science workflows.
- Detailed breakdown: Selecting the right visualization for categorical, numerical, and time-series variables.
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.